

Mock Semestral — Solutions

Parametric Inference, B.Stat. 3rd Year, Semester 1, 2026–27

Marks in **[green]** are indicative; total 100.

Do not read this before attempting the paper under timed conditions.

What each question tests

Q	Marks	Topic	Source in the course
1	12	Factorisation, Rao–Blackwell, incompleteness	Lec 2–5
2	12	Completeness, UMVUE, polynomial estimability	Lec 2–4; HW1 Q1
3	12	Rao–Blackwellisation, CRLB non-attainment	Lec 2–4; HW1 Q2–Q3
4	10	Multiparameter completeness, ancillarity, Basu	Lec 4–5; HW1 Q5–Q6
5	12	Conjugacy, Bayes estimate, improper priors	Lec 6–7; HW2 Q3
6	12	Bayes risk, equalizer rules, minimax	Lec 8; HW2 Q1–Q2
7	12	Neyman–Pearson; failure of two-sided UMP	Lec 6, 9; HW2 Q5, HW3 Q3
8	10	MLR and Karlin–Rubin	Lec 9; HW3 Q1–Q2, Q4
9	10	Consistency; Cauchy pathology; regression	Lec 10

Self-assessment. ≥ 80 : you are in good shape — spend the remaining time on Q3(c), Q6(b) and Q7(c), which are the discriminating parts. $60\text{--}80$: re-derive the exponential-family completeness criterion and the Karlin–Rubin statement until they are automatic. < 60 : work Part II of `PI_Class_Exercises_Solved.pdf` before anything else.

Question 1 (12 marks — bookwork)

(a) **[3]** *Neyman–Fisher factorisation.* Let $X_1, \dots, X_n$ have joint pdf/pmf $f(\mathbf{x} \mid \theta)$. A statistic $S(\mathbf{X})$ is sufficient for θ if and only if there exist functions g and $h \geq 0$ with

$$f(\mathbf{x} \mid \theta) = g(\theta, S(\mathbf{x})) \cdot h(\mathbf{x}) \quad \text{for all } \mathbf{x}, \theta,$$

where h does not involve θ and the dependence on $\mathbf{x}$ in g is only through $S(\mathbf{x})$.

(b) **[6]** *Rao–Blackwell.* Let T be an estimator of $\psi(\theta)$ with $\mathbb{E}_\theta T^2 < \infty$, and let S be sufficient for θ . Put $T^* = \mathbb{E}[T \mid S]$. Then T^* is a statistic, $\mathbb{E}_\theta T^* = \mathbb{E}_\theta T$, and $R(\theta, T^*) \leq R(\theta, T)$ for every θ .

Proof. T^* is a genuine statistic because sufficiency of S makes the conditional distribution of $\mathbf{X}$ given S , and hence $\mathbb{E}[T \mid S]$, free of θ **[1]**. By the tower property,

$$\mathbb{E}_\theta T^* = \mathbb{E}_\theta [\mathbb{E}(T \mid S)] = \mathbb{E}_\theta T,$$

so T and T^* have the *same bias* **[1]**. By the conditional variance decomposition,

$$\text{Var}_\theta(T) = \mathbb{E}_\theta [\text{Var}(T \mid S)] + \text{Var}_\theta (\mathbb{E}[T \mid S]) \geq \text{Var}_\theta(T^*),$$

since the first term is non-negative **[2]**. Adding the common squared bias,

$$R(\theta, T) = \text{Bias}^2 + \text{Var}_\theta(T) \geq \text{Bias}^2 + \text{Var}_\theta(T^*) = R(\theta, T^*). \quad \square \text{ [1]}$$

Strictness **[1]**: the inequality is strict at θ iff $\mathbb{E}_\theta [\text{Var}(T \mid S)] > 0$, i.e. iff T is not already (a.s.) a function of S .

(c) [3] Take $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}[\theta, \theta + 1]$, $\theta \in \mathbb{R}$. By factorisation $S = (X_{(1)}, X_{(n)})$ is sufficient. Writing $X_i = \theta + U_i$ with $U_i \stackrel{\text{iid}}{\sim} U[0, 1]$, the range $X_{(n)} - X_{(1)} = U_{(n)} - U_{(1)}$ has a distribution free of θ — it is $\text{Beta}(n-1, 2)$, with mean $\frac{n-1}{n+1}$. Hence

$$g(u, v) := (v - u) - \frac{n-1}{n+1} \quad \text{satisfies} \quad \mathbb{E}_\theta g(X_{(1)}, X_{(n)}) = 0 \quad \forall \theta,$$

while $g \neq 0$. So S is sufficient but not complete.

(Equally acceptable: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta)$, where the order statistic is sufficient and $g(X_{(i)} - X_{(j)})$ gives the same obstruction; or $N(\theta, \theta^2)$, whose natural parameter space is a curve, not a rectangle.)

Question 2 (12 marks — Binomial: completeness, UMVUE, estimability)

(a) [4]

$$f(x | \theta) = \binom{m}{x} \theta^x (1 - \theta)^{m-x} = \underbrace{(1 - \theta)^m}_{p(\theta)} \underbrace{\binom{m}{x}}_{h(x)} \exp \left\{ \underbrace{\log \frac{\theta}{1-\theta}}_{c(\theta)} \cdot \underbrace{x}_{T(x)} \right\},$$

a one-parameter exponential family with $T(x) = x$ [2]. As θ ranges over $(0, 1)$, the natural parameter $\eta = \log \frac{\theta}{1-\theta}$ ranges over all of $\mathbb{R}$, which certainly contains an interval with non-empty interior; the support $\{0, 1, \dots, m\}$ is free of θ . By the completeness criterion for exponential families, X is complete sufficient [2].

Direct verification, if preferred. $\mathbb{E}_\theta g(X) = 0 \quad \forall \theta$ gives, after dividing by $(1 - \theta)^m$ and setting $r = \theta/(1 - \theta) \in (0, \infty)$, $\sum_{x=0}^m g(x) \binom{m}{x} r^x = 0$ for all $r > 0$; a polynomial vanishing on an interval has all coefficients zero, so $g \equiv 0$.

(b) [5] Use factorial moments. For $1 \leq k \leq m$, with $X^k = X(X-1) \cdots (X-k+1)$,

$$\mathbb{E}_\theta [X^k] = \sum_{x=k}^m \frac{m!}{(x-k)!(m-x)!} \theta^x (1 - \theta)^{m-x} = m^k \theta^k \sum_{y=0}^{m-k} \binom{m-k}{y} \theta^y (1 - \theta)^{m-k-y} = m^k \theta^k,$$

substituting $y = x - k$ and recognising the $\text{Bin}(m-k, \theta)$ total [3]. Hence

$$\widehat{\theta^k} = \frac{X(X-1) \cdots (X-k+1)}{m(m-1) \cdots (m-k+1)}$$

is unbiased, and being a function of the complete sufficient statistic X it is the UMVUE by Lehmann–Scheffé [2].

Check $k = 1$: X/m . Check $k = 2$: $X(X-1)/\{m(m-1)\}$, which combined with $k = 1$ gives the familiar UMVUE $\frac{X(m-X)}{m(m-1)}$ of $\theta(1 - \theta)$.

(c) [3] For any estimator T ,

$$\mathbb{E}_\theta T = \sum_{x=0}^m T(x) \binom{m}{x} \theta^x (1 - \theta)^{m-x}$$

is a polynomial in θ of degree at most m [2]. If it equalled θ^{m+1} for every $\theta \in (0, 1)$, then two polynomials would agree on an interval and hence be identical — but their degrees are $\leq m$ and $m+1$. Contradiction [1]. So no unbiased estimator of θ^{m+1} exists.

Where marks are lost

In (c) it is not enough to say “ θ^{m+1} has degree $m+1$ ”. You must say *why* agreement on an interval forces equality of coefficients — a polynomial with infinitely many roots is

identically zero. The same one-line principle is what makes $1/\theta$ non-estimable.

Question 3 (12 marks — Poisson: UMVUE of $e^{-\lambda}$ and the CRLB)

(a) [6] $S = \sum_i X_i \sim \text{Poisson}(n\lambda)$ is complete sufficient (exponential family, $T(x) = x$, $c(\lambda) = \log \lambda$, natural parameter space $\mathbb{R}$) [1].

Route B (*Rao-Blackwell*). $T_0 = \mathbb{I}\{X_1 = 0\}$ is unbiased for $e^{-\lambda}$, so the UMVUE is $\mathbb{E}[T_0 | S] = \mathbb{P}(X_1 = 0 | S = s)$. Given $S = s$, the conditional law of X_1 is $\text{Bin}(s, \frac{1}{n})$ (from the multinomial conditional distribution of Lecture 2, Ex 5), so

$$\mathbb{P}(X_1 = 0 | S = s) = \left(1 - \frac{1}{n}\right)^s = \left(\frac{n-1}{n}\right)^s. \quad [4]$$

Being unbiased and a function of the complete sufficient statistic, by Lehmann–Scheffé

$$\widehat{e^{-\lambda}} = \left(\frac{n-1}{n}\right)^S, \quad S = \sum_{i=1}^n X_i. \quad [1]$$

Route A (*series matching*), equally acceptable. Require $\mathbb{E}_\lambda g(S) = e^{-\lambda}$:

$$\sum_{s \geq 0} g(s) \frac{e^{-n\lambda} (n\lambda)^s}{s!} = e^{-\lambda} \iff \sum_{s \geq 0} \frac{g(s) n^s}{s!} \lambda^s = e^{(n-1)\lambda} = \sum_{s \geq 0} \frac{(n-1)^s}{s!} \lambda^s,$$

so $g(s)n^s = (n-1)^s$, giving the same answer.

(b) [3] For $S \sim \text{Poisson}(n\lambda)$ and any $a > 0$, $\mathbb{E}[a^S] = e^{-n\lambda} \sum_s \frac{(an\lambda)^s}{s!} = e^{n\lambda(a-1)}$ [1]. With $a = \frac{n-1}{n}$ this gives $e^{-\lambda}$, confirming unbiasedness. With $a = \left(\frac{n-1}{n}\right)^2$,

$$n\lambda(a-1) = n\lambda \cdot \frac{(n-1)^2 - n^2}{n^2} = \frac{\lambda(1-2n)}{n} = -2\lambda + \frac{\lambda}{n},$$

so $\mathbb{E}[a^S] = e^{-2\lambda} e^{\lambda/n}$ and

$$\text{Var}_\lambda \left(\left(\frac{n-1}{n}\right)^S \right) = e^{-2\lambda} (e^{\lambda/n} - 1). \quad [2]$$

(c) [3] With $\psi(\lambda) = e^{-\lambda}$ we have $\psi'(\lambda) = -e^{-\lambda}$, and for the Poisson $I(\lambda) = 1/\lambda$, so

$$\text{CRLB} = \frac{[\psi'(\lambda)]^2}{n I(\lambda)} = \frac{\lambda e^{-2\lambda}}{n}. \quad [1]$$

Since $e^u - 1 > u$ for every $u > 0$, with $u = \lambda/n$,

$$\text{Var} = e^{-2\lambda} (e^{\lambda/n} - 1) > e^{-2\lambda} \cdot \frac{\lambda}{n} = \text{CRLB}. \quad [1]$$

So the bound is **not attained**, at any λ or n . This is consistent with the attainment criterion: equality requires the score to factor as $k(\lambda)[T - \psi(\lambda)]$, i.e. ψ must be the mean of the natural sufficient statistic — here $\psi(\lambda) = e^{-\lambda}$ is a non-linear function of λ , so it cannot be.

As $n \rightarrow \infty$ [1]: $e^{\lambda/n} - 1 = \frac{\lambda}{n} + O(n^{-2})$, so the ratio $\text{Var}/\text{CRLB} \rightarrow 1$. The UMVUE is asymptotically efficient even though it never attains the bound exactly.

Where marks are lost

“Does not attain the CRLB” does *not* contradict being the UMVUE. The CRLB is a lower bound on the variance of *all* unbiased estimators; if no estimator attains it, the best one still sits above it. Writing “since the variance exceeds the CRLB, it is not the UMVUE” is the single most common error on this type of question.

Question 4 (10 marks — Basu applied to the sample range)

(a) [4]

$$f(\mathbf{x} \mid \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_i x_i^2 + \frac{\mu}{\sigma^2} \sum_i x_i - \frac{n\mu^2}{2\sigma^2} \right\},$$

a two-parameter exponential family with $T = (\sum X_i^2, \sum X_i)$ and natural parameters

$$(\eta_1, \eta_2) = \left(-\frac{1}{2\sigma^2}, \frac{\mu}{\sigma^2} \right) \quad [2].$$

As (μ, σ^2) ranges over $\mathbb{R} \times (0, \infty)$, (η_1, η_2) ranges over the open half-plane $(-\infty, 0) \times \mathbb{R}$, which contains a two-dimensional rectangle with non-empty interior. By the multiparameter completeness criterion, T is complete sufficient [2]. (Equivalently $(\bar{X}, S^2)$, a one-to-one function of T .)

(b) [3] Write $X_i = \mu + \sigma Z_i$ with $Z_i \stackrel{\text{iid}}{\sim} N(0, 1)$. Order statistics are monotone equivariant and $\sigma > 0$, so $X_{(k)} = \mu + \sigma Z_{(k)}$; hence

$$R = X_{(n)} - X_{(1)} = \sigma(Z_{(n)} - Z_{(1)}), \quad S = \sigma S_Z \quad \text{where } S_Z^2 = \frac{1}{n-1} \sum_i (Z_i - \bar{Z})^2.$$

Therefore

$$V = \frac{R}{S} = \frac{Z_{(n)} - Z_{(1)}}{S_Z},$$

whose distribution involves neither μ (it cancelled in the differences) nor σ (it cancelled in the ratio). So V is **ancillary**.

(c) [3] $(\bar{X}, S^2)$ is complete sufficient by (a) and V is ancillary by (b), so Basu's theorem gives $V \perp\!\!\!\perp (\bar{X}, S^2)$; in particular $V \perp\!\!\!\perp S$ [2]. Since $R = VS$,

$$\mathbb{E}[R] = \mathbb{E}[VS] = \mathbb{E}[V] \mathbb{E}[S]. \quad [1]$$

(This is exactly the classical identity behind tabulated “ d_2 ” constants: $\mathbb{E}[R] = d_2\sigma$ with $d_2 = \mathbb{E}[V] \mathbb{E}[S]/\sigma$ a pure function of n .)

Where marks are lost

V is ancillary for the *pair* (μ, σ^2) — both must cancel. A common slip is to check only the location cancellation and forget the scale, which is what the division by S (rather than by a constant) achieves.

Question 5 (12 marks — conjugate and generalized Bayes for the exponential scale)

(a) [5] The likelihood is $L(\theta) = \theta^{-n} \exp\{-S/\theta\}$ with $S = \sum_i x_i$ [1]. Hence

$$\pi(\theta \mid \mathbf{x}) \propto \theta^{-n} e^{-S/\theta} \cdot \theta^{-(\alpha+1)} e^{-\beta/\theta} = \theta^{-(\alpha+n+1)} \exp \left\{ -\frac{\beta + S}{\theta} \right\} \quad [3],$$

which is again an inverse-gamma kernel. So the family is conjugate and

$$\theta \mid \mathbf{X} \sim \text{Inverse-Gamma}(\alpha + n, \beta + S). \quad [1]$$

(b) [3] An Inverse-Gamma(a, b) variable has mean $b/(a-1)$, finite iff $a > 1$ [1]. Hence the Bayes estimate under squared error is the posterior mean

$$T_{\alpha, \beta} = \frac{\beta + \sum_i X_i}{\alpha + n - 1} \quad \text{provided } \alpha + n > 1,$$

which holds for every $n \geq 1$ since $\alpha > 0$ [2].

(c) [2] $\mathbb{E}_\theta [\sum_i X_i] = n\theta$, so

$$\mathbb{E}_\theta T_{\alpha, \beta} = \frac{\beta + n\theta}{\alpha + n - 1}.$$

For this to equal θ for *all* $\theta > 0$ we would need $\beta = 0$ and $n = \alpha + n - 1$, i.e. $\alpha = 1$, $\beta = 0$ — which is not a proper prior. So for every proper inverse-gamma prior the Bayes estimate is **biased**. This is an instance of the general fact that a Bayes estimate is unbiased only in degenerate problems.

(d) [2] $\pi(\theta) = 1/\theta$ is the $\alpha = \beta = 0$ member of the family (improper, since $\int_0^\infty \theta^{-1} d\theta = \infty$). The posterior kernel is $\theta^{-(n+1)} e^{-S/\theta}$, i.e. Inverse-Gamma(n, S), which is a *proper* density. Its mean exists for $n \geq 2$, giving the generalized Bayes estimate

$$\hat{\theta} = \frac{\sum_i X_i}{n-1} = \frac{n}{n-1} \bar{X} \quad (n \geq 2).$$

This is *not* $\bar{X}$: it inflates the UMVUE by the factor $n/(n-1)$, so $\mathbb{E}_\theta \hat{\theta} = \frac{n\theta}{n-1} > \theta$ — biased upwards, though the bias is $O(1/n)$ and vanishes asymptotically. As always, the generalized Bayes estimate is a function of the sufficient statistic $\sum X_i$.

Question 6 (12 marks — the minimax Binomial estimator)

(a) [3] $\pi(\theta | x) \propto \theta^x (1-\theta)^{n-x} \cdot \theta^{p-1} (1-\theta)^{q-1} = \theta^{p+x-1} (1-\theta)^{q+n-x-1}$, the kernel of Beta($p+x, q+n-x$) [2]. A Beta(a, b) has mean $a/(a+b)$, so the Bayes estimate is

$$T_{p,q} = \mathbb{E}[\theta | X] = \frac{p+X}{(p+X) + (q+n-X)} = \frac{p+X}{p+q+n}. \quad [1]$$

(b) [6] Write $c = p+q+n$. Since $\mathbb{E}_\theta X = n\theta$ and $\text{Var}_\theta X = n\theta(1-\theta)$,

$$\text{Bias} = \frac{p+n\theta}{c} - \theta = \frac{p - (p+q)\theta}{c}, \quad \text{Var}_\theta(T_{p,q}) = \frac{n\theta(1-\theta)}{c^2} \quad [2],$$

so

$$R(\theta, T_{p,q}) = \frac{1}{c^2} \left[n\theta(1-\theta) + (p - (p+q)\theta)^2 \right] = \frac{1}{c^2} \left[\underbrace{((p+q)^2 - n)}_{\theta^2} \theta^2 + \underbrace{(n - 2p(p+q))}_{\theta} \theta + p^2 \right] \quad [2].$$

The risk is free of θ iff both bracketed coefficients vanish:

$$(p+q)^2 = n \implies p+q = \sqrt{n}, \quad n = 2p(p+q) = 2p\sqrt{n} \implies p = \frac{\sqrt{n}}{2}, \quad q = \frac{\sqrt{n}}{2} \quad [2].$$

This is the *only* such pair. Then $c = n + \sqrt{n} = \sqrt{n}(1 + \sqrt{n})$ and

$$R(\theta, T^*) = \frac{p^2}{c^2} = \frac{n/4}{n(1 + \sqrt{n})^2} = \boxed{\frac{1}{4(1 + \sqrt{n})^2}} \quad \text{for every } \theta.$$

(c) [3] $T^* = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}$ is Bayes with respect to the proper prior Beta($\frac{\sqrt{n}}{2}, \frac{\sqrt{n}}{2}$) and has constant risk. By the constant-risk criterion — for any T ,

$$\sup_\theta R(\theta, T) \geq r(\pi, T) \geq r(\pi, T^*) = \frac{1}{4(1 + \sqrt{n})^2} = \sup_\theta R(\theta, T^*)$$

— T^* is **minimax** [2], and $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ is least favourable.

The UMVUE X/n has $R(\theta, X/n) = \theta(1 - \theta)/n$, maximised at $\theta = \frac{1}{2}$:

$$\max_{\theta} R(\theta, X/n) = \frac{1}{4n} > \frac{1}{4(1 + \sqrt{n})^2} \quad \text{since } (1 + \sqrt{n})^2 = n + 2\sqrt{n} + 1 > n.$$

So the UMVUE is **not** minimax [1]; but the ratio $\frac{(1+\sqrt{n})^2}{n} \rightarrow 1$, so the advantage disappears for large n . (Note also that T^* beats X/n only near $\theta = \frac{1}{2}$ — near the endpoints X/n has much smaller risk. Minimaxity is a worst-case criterion, not a uniform one.)

Question 7 (12 marks — Neyman–Pearson, and why two-sided UMP fails)

(a) [6] Both hypotheses are simple, so Neyman–Pearson applies. With $f_0 = \varphi(x)$ and $f_1(x) = \frac{1}{2\sqrt{2\pi}} e^{-x^2/8}$,

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)} = \frac{\frac{1}{2} e^{-x^2/8}}{e^{-x^2/2}} = \frac{1}{2} \exp \left\{ x^2 \left(\frac{1}{2} - \frac{1}{8} \right) \right\} = \frac{1}{2} e^{3x^2/8} \quad [3].$$

Λ is a strictly increasing function of x^2 , so $\{\Lambda > k\} = \{x^2 > k'\} = \{|x| > c\}$ [2]. Both distributions are continuous, so no randomisation is needed and c is fixed by the size:

$$\text{Reject } H_0 \iff |X| > c, \quad \mathbb{P}_{H_0}(|X| > c) = 2\{1 - \Phi(c)\} = \alpha, \quad \text{i.e. } c = z_{\alpha/2}. \quad [1]$$

(b) [3] For $\alpha = 0.05$, $c = z_{0.025} = 1.960$: reject iff $|X| > 1.96$ [1]. Under H_A , $X \sim N(0, 4)$ so $X/2 \sim N(0, 1)$ and

$$\text{power} = \mathbb{P}(|X| > 1.96) = 2\left\{1 - \Phi\left(\frac{1.96}{2}\right)\right\} = 2\{1 - \Phi(0.98)\} = 2(1 - 0.8365) = \boxed{0.327} \quad [2].$$

(Only 0.327 against a variance four times too large: a single observation carries very little information about a scale parameter.)

(c) [3] **No UMP test exists.** The family $N(0, \sigma^2)$ is a one-parameter exponential family in $T(x) = x^2$ with $c(\sigma^2) = -\frac{1}{2\sigma^2}$, increasing in σ^2 , so it has MLR in X^2 [1]. Consequently:

- Against any $\sigma_1^2 > 1$, the same computation as in (a) gives $\Lambda(x) = \frac{1}{\sigma_1} \exp\{\frac{x^2}{2}(1 - \sigma_1^{-2})\}$, increasing in x^2 , so the MP test rejects for *large* $|X|$ [1].
- Against any $\sigma_1^2 < 1$, the coefficient $\frac{1}{2}(1 - \sigma_1^{-2})$ is *negative*, so Λ is decreasing in x^2 and the MP test rejects for *small* $|X|$ [1].

The two families of MP tests are disjoint (a level- α test cannot reject both the largest and the smallest $|x|$ values), so no single test of size α is simultaneously most powerful against every $\sigma^2 \neq 1$. Hence no UMP test exists, and one restricts attention to unbiased tests (UMPU), where the two-sided χ^2 test with suitably chosen — not equal-tailed — cutoffs is optimal.

Where marks are lost

The contrast worth remembering: the failure here is the *generic* one (MLR family, two-sided alternative). It is not contradicted by HW3 Q4, where a two-sided UMP test *does* exist for $U[0, \theta]$ — there the support moves with θ , so rejecting on $\{X > \theta_0\}$ costs nothing under H_0 and both sides can be served at once.

Question 8 (10 marks — MLR and Karlin–Rubin for the uniform)

(a) [4] The joint density is $f_{\theta}(\mathbf{x}) = \theta^{-n} \mathbb{I}[0 \leq x_{(1)}] \mathbb{I}[x_{(n)} \leq \theta]$, so $X_{(n)}$ is sufficient. For $\theta_0 < \theta_1$,

$$\frac{f_{\theta_1}(\mathbf{x})}{f_{\theta_0}(\mathbf{x})} = \left(\frac{\theta_0}{\theta_1}\right)^n \cdot \frac{\mathbb{I}[x_{(n)} \leq \theta_1]}{\mathbb{I}[x_{(n)} \leq \theta_0]} = \begin{cases} (\theta_0/\theta_1)^n, & x_{(n)} \leq \theta_0, \\ +\infty, & \theta_0 < x_{(n)} \leq \theta_1, \end{cases}$$

(the region $x_{(n)} > \theta_1$ has probability zero under both) [3]. As a function of $x_{(n)}$ this takes the constant value $(\theta_0/\theta_1)^n < 1$ and then jumps to $+\infty$: it is **non-decreasing in** $x_{(n)}$, so the family has MLR in $X_{(n)}$ [1].

(b) [4] By Karlin–Rubin the UMP level- α test of $H_0 : \theta \leq \theta_0$ against $H_A : \theta > \theta_0$ rejects for large $X_{(n)}$ [1]. Under θ_0 , $X_{(n)}$ has cdf $(x/\theta_0)^n$ on $[0, \theta_0]$, so with rejection region $\{X_{(n)} > c\}$,

$$\mathbb{P}_{\theta_0}(X_{(n)} > c) = 1 - \left(\frac{c}{\theta_0}\right)^n = \alpha \implies c = \theta_0(1 - \alpha)^{1/n} \text{ [2].}$$

$X_{(n)}$ is continuous, so no randomisation is needed:

$$\text{Reject } H_0 \iff X_{(n)} > \theta_0(1 - \alpha)^{1/n}. \text{ [1]}$$

(c) [2] With $c = \theta_0(1 - \alpha)^{1/n}$,

$$\beta(\theta) = \mathbb{P}_{\theta}(X_{(n)} > c) = \begin{cases} 0, & \theta \leq c, \\ 1 - \frac{(1 - \alpha)\theta_0^n}{\theta^n}, & \theta > c. \end{cases}$$

β is non-decreasing in θ , with $\beta(\theta_0) = 1 - (1 - \alpha) = \alpha$ and $\beta(\theta) \rightarrow 1$ as $\theta \rightarrow \infty$. Monotonicity gives $\sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0) = \alpha$, so the test has size exactly α over the composite null.

Where marks are lost

Note that $\{X_{(n)} > \theta_0\}$ is *contained* in the rejection region, since $c < \theta_0$. That is forced: any observation exceeding θ_0 is impossible under H_0 , so an optimal test must reject there. Answers that take the rejection region to be $\{X_{(n)} > \theta_0\}$ alone have size 0, not α , and are not UMP.

Question 9 (10 marks — consistency)

(a) [6] $\bar{X}$ is not consistent. The Cauchy family is *stable*: if $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta, 1)$ then $\sum_i X_i \sim \text{Cauchy}(n\theta, n)$, and hence

$$\bar{X} \sim \text{Cauchy}(\theta, 1) \quad \text{for every } n \text{ [2].}$$

(Quickest justification: the characteristic function of $\text{Cauchy}(\theta, 1)$ is $e^{i\theta t - |t|}$, so that of $\bar{X}$ is $(e^{i\theta t/n - |t|/n})^n = e^{i\theta t - |t|}$.) Therefore, for any $\varepsilon > 0$,

$$\mathbb{P}_{\theta}(|\bar{X} - \theta| > \varepsilon) = 1 - \frac{2}{\pi} \arctan \varepsilon,$$

a strictly positive constant that does not depend on n [1]. So $\bar{X} \not\stackrel{p}{\rightarrow} \theta$: averaging buys nothing at all. (Consistent with $\mathbb{E}|X_1| = \infty$, so the weak law does not apply.)

The sample median is consistent. Let F be the $\text{Cauchy}(\theta)$ cdf, so $F(\theta) = \frac{1}{2}$ and F is continuous and strictly increasing. Let $\tilde{X}_n$ be the sample median. For $\varepsilon > 0$,

$$\mathbb{P}(\tilde{X}_n - \theta > \varepsilon) \leq \mathbb{P}(\text{at least } \frac{n}{2} \text{ of the } X_i \text{ exceed } \theta + \varepsilon) = \mathbb{P}\left(\text{Bin}(n, 1 - F(\theta + \varepsilon)) \geq \frac{n}{2}\right),$$

and $1 - F(\theta + \varepsilon) < \frac{1}{2}$ strictly, so by the weak law (or Chebyshev) this probability $\rightarrow 0$ [2]. Symmetrically $\mathbb{P}(\tilde{X}_n - \theta < -\varepsilon) = \mathbb{P}(\text{Bin}(n, F(\theta - \varepsilon)) \geq \frac{n}{2}) \rightarrow 0$ since $F(\theta - \varepsilon) < \frac{1}{2}$ [1]. Hence $\tilde{X}_n \xrightarrow{p} \theta$.

(b) [4] With $S_{xx} = \sum_i (x_i - \bar{x})^2$,

$$\hat{\beta}_{LS} = \frac{\sum_i (x_i - \bar{x})Y_i}{S_{xx}}, \quad \mathbb{E}\hat{\beta}_{LS} = \frac{\sum_i (x_i - \bar{x})(\alpha + \beta x_i)}{S_{xx}} = \beta, \quad \text{Var}(\hat{\beta}_{LS}) = \frac{\sigma^2}{S_{xx}} \text{ [2]}$$

(using $\sum_i (x_i - \bar{x}) = 0$ for the first and independence for the second). The estimator is unbiased, so by Chebyshev

$$\mathbb{P}(|\hat{\beta}_{LS} - \beta| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2 S_{xx}}.$$

Hence the condition is

$$\sum_{i=1}^n (x_i - \bar{x})^2 \longrightarrow \infty \text{ as } n \rightarrow \infty \quad [2]$$

i.e. **the design points must be sufficiently spread out**. Merely taking $n \rightarrow \infty$ is not enough: if all x_i were equal, $S_{xx} = 0$ and β is not even identifiable; if the x_i cluster fast enough that S_{xx} stays bounded, the variance does not vanish and $\hat{\beta}_{LS}$ is inconsistent however large n becomes.